

AE4262 Assignment

Instructions:

1. *Prepare a report on the assignment.*
2. *The report should contain at least the following parts*
 - *Problem description*
 - *Assumptions*
 - *Procedure*
 - *Results & observations*
 - *Conclusions & outlook*
3. *The report should be handed in **before 10th June 23:59 CET***
4. *You can do this assignment in a group of three/four students.*
5. *Please submit your assignment on Brightspace. If in a group, you still have to submit individually and indicate clearly name and student number of all authors at the beginning of your report.*
6. *Grading: the score in brackets represents the percentage of the grade that is assigned to each section. An additional 10% is assigned to the quality of the report.*
7. *Each team member is expected to write the full report and be fully aware of the topics discussed in it. If the instructors suspect that you copied the report without taking active part in it, your report might be invalidated and students would be reported to the board of examiners.*
8. *Where not specified, points are equally-distributed among subsections. Also, 10% of the total grade of this assignment is awarded for clarity, conciseness and presentation quality.*

CEA instructions: After downloading and unpacking the software, double-click on CEA executable (cea-exec). A subfolder will be created. Copy and past the input files on Brightspace within this folder. Edit the file as required by your exercise and save. Then run FCEA2m. A prompt will open: insert the exact name of your input file and press Return. An output file and an excel file will be created, containing the results of the computation.

CANTERA instructions: CANTERA is available in several programming languages, though the Python interface is recommended. Please note that the latest version (3.2) is not available in MATLAB. To install Cantera, use either Conda or Pip:

- `pip install cantera`
- `conda create --name ct-env --channel conda-forge cantera ipython matplotlib jupyter` *#Creates a Conda environment with CANTERA*
`conda activate ct-env` *#Activates the environment*

ipython, matplotlib and jupyter are optional. Python 3.10 or greater is required.

For further information refer to software's instructions (links provided on Brightspace and the tutorial's slides).

Experimental part

Please refer to the document “Laboratory Exercise_2026” for the description of the setup

Part 1 [5% of the final mark]

Flow calculations

- Write down and balance the global chemical reactions for methane-air and hydrogen-air combustion at stoichiometric, lean, and rich conditions, using a generic equivalence ratio ϕ .
- For each methane-air flames and hydrogen-air flames, show how the required fuel and air flow rates, flow velocities, and Reynolds numbers were determined. You should clearly show how the quantities needed for the Reynolds number calculation were obtained. Clearly state all assumptions made. Present your test matrices in a table.

Part 2 [15% of the final mark]

Flame speed calculations

- Calculate the laminar flame speed of the H_2 /air and CH_4 /air mixtures using the Bunsen flame angle method applied to the OH-PLIF images. Plot the measured flame speeds as a function of equivalence ratio. In the same plot, compare the experimental values with unstretched laminar flame speeds obtained from literature and from one-dimensional freely propagating flame simulations using Cantera. Discuss why the maximum flame speed does not occur at $\phi = 1$.
- Comment on any deviations between the experimental and theoretical values. Discuss possible sources of uncertainty in the flame-speed measurements and suggest improvements to the experiment and methodology to reduce errors.
- Discuss why the flame shape deviates from a perfect cone near the burner rim and at the flame tip.
- Explain why the hydrogen Bunsen burner had a smaller diameter than the methane Bunsen burner.

Part 3 [15% of the final mark]

Blowoff and flashback

- Discuss the physical mechanisms leading to blow-off and flashback in the hydrogen-air and methane-air premixed Bunsen flames considered in this assignment. Your discussion should consider flame speed, bulk flow velocity, equivalence ratio, heat losses, and boundary-layer effects.
- For the experimental burner geometries and thermal powers used in the experiment, estimate the fuel and air flow rates at which flashback is expected to occur. Compare these predicted values with the experimentally observed onset of flashback, and discuss possible reasons for any differences.

Part 4 [10% of the final mark]

Diagnostics

- Explain the differences in the flame emission spectra measured for the three premixed methane-air flames at different equivalence ratios ($\phi=0.8, 1, 1.2$) and identify the main chemical species responsible for the observed spectral emissions.
- Sketch the flame emission spectra expected for premixed hydrogen-air flames and diffusion methane flames. Use a graph with wavelength (200–800 nm) on the x-axis and relative emission intensity on the y-axis. Identify the main spectral features and emitting species expected for each flame type.
- For hydrogen-air flames, apply an Abel inversion to the OH* chemiluminescence images and compare the reconstructed fields with the OH-PLIF images for $\phi=0.7, 1, 1.3$. Discuss the differences between the OH* chemiluminescence and OH-PLIF measurements.

[Hint: you may use a one-dimensional flame simulation to compare the spatial profiles of OH concentration and heat release rate. Based on these profiles, where do you expect the OH* chemiluminescence signal to peak?]

Numerical part

Please refer to the tutorial slides on Brightspace for information on thermodynamics databases and use of the software. Sample exercises are also available on Brightspace.

Part 1 [15% of the final mark]

- Calculate and plot the variation of c_p for H_2 , O_2 , N_2 , CO_2 , H_2O and CH_4 from 300 to 2400K.
- The C_p for the gas mixture can then be calculated by using the relation

$$c_{p,mix} = \sum_{k=1}^N Y_k c_{p,k}$$

Calculate and plot (in the same figure) the variation of $c_{p,mix}$ for reactants and combustion products from 300 to 2400K, assuming the reactants to be air and fuel at equivalence ratios 0.5, 0.8, 1.0, 1.2, 1.4. Repeat the computations using hydrogen and methane as fuel. Assume complete combustion. What are your observations in terms of C_p variation (max 200 words)?

- Compute and plot the flame adiabatic temperature for complete combustion of H_2 /air and reactant temperature of 1100K for the same equivalence ratios. Compare this with the adiabatic flame temperature in the case of methane/air combustion.

What are your key observations and conclusions? Where do differences come from? (max 200 words)

Part 2 [20% of the final mark]

Chemical equilibrium (Using CANTERA or CEA)

Note: Use always baseline conditions unless indicated otherwise. Baseline conditions are indicated in **bold**

Note: Calculations have to be done at constant pressure and enthalpy (sample input files for NASA CEA are available on Brightspace)

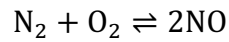
Compute the following using hydrogen as fuel and then repeat the calculations using methane as fuel. Report the results for methane and hydrogen for each point below in the same graph.

- Evaluate the variation in adiabatic flame temperature with equivalence ratio for $\phi = 0.5, 0.6, 0.7, \mathbf{0.8}, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4$.
- Evaluate the variation in adiabatic flame temperature with Pressure, for **1**, 10, and 40 bar.
- Evaluate the variation in adiabatic flame temperature with air inlet temperature at **300K**, 600K, 1100K.
- Evaluate the NO emissions (in ppm) at 300K and 1bar for the range of ϕ in point a)

Part 3 [10% of the final mark]

Estimating NO_x

- a) [4 pts] Consider a two-steps reaction at 1 bar, where the first step is the global reaction of fuel/air combustion for $T_{\text{reactants}} = 300\text{K}$, and the second is the following equilibrium reaction:



Repeat the computation for both methane and hydrogen as fuel and for three equivalence ratios: $\phi = 0.5, 0.8$ and 1.0 . Plot the NO mass fraction as function of equivalence ratio.

Note: Only compute the composition in the first step. Use the equilibrium temperature from Part 2 for the second step.

- b) [6 pts] Now compare the results with those you computed numerically for both 1D flames and equilibrium, and comment on the differences. In particular explain in three separate bullet points (of max 50 words each)
- i) the difference between your NO computation and the one from equilibrium with CANTERA or CEA at lean conditions;
 - ii) the same at the stoichiometric condition; and
 - iii) the difference in NO obtained using chemical equilibrium (from CANTERA or CEA) and 1D flames.